Complete Segal Objects

Nima Rasekh

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The goal of this talk is to discuss internal higher categories. We first review category objects in the classical setting and try to generalize that to higher categories by employing the language of complete Segal objects. Finally we use Cartesian fibrations to construct a presheaf valued in higher categories from each complete Segal object.

Category Objects

Let us first shortly recall the 1-categorical case. If $\mathcal{C}$ is a category with finite limits then we can define category objects as follows:

Definition 1.1. A category object interal to $\mathcal{C} = (O, M)$ is the following data:

1. **Objects:** An object $O \in \mathcal{C}$ of "objects" and a an object $M \in \mathcal{C}$ of morphisms.
2. **Morphisms:** A map $(s,t)M \to O \times O$ giving a "source" and "target".
3. **Identity:** A section of both s and t, $i : O \to M$ which is the "identity map"
4. **Composition**: A map $\mu : M^t \times_O^s M \to M$ a composition map
5. **Composing with Identity:**The following diagrams have to commute (Composition with identity)

$$\begin{tikzcd} M \arrow[r, "(i\circ t, id)"] \arrow[dr, "id"'] & M \times M \arrow[d, "\mu"] & M \arrow[l, "(id, i\circ s)"'] \arrow[dl, "id"] \\ & M & \end{tikzcd}$$

6. **Associativity:** Following diagram commutes:

$$\begin{tikzcd} M \times_O M \times_O M \arrow[r, "(id,\mu)"] \arrow[d, "(\mu,id)"] & M \times_O M \arrow[d, "\mu"] \\ M \times_O M \arrow[r, "\mu"] & M \end{tikzcd}$$

Remark 1.2. Note that the category does not necessarily needs to have all finite limits. It suffices to have enough to be able to construct couple products and the pullback object $M \times_O M$.

Let us see some examples:

Example 1.3. A category object internal to sets is just a category itself.

Example 1.4. A category object interal to categories is sometimes called a "double category"

Example 1.5. Let $\mathcal{C}$ be any category and $\mathcal{G}$ be a group object in $\mathcal{C}$. Then in particular $\mathcal{G}$ is a category object, where O is just the final object, $*$, and M is the group object itself.

Example 1.6. As an example of the previous part, topological groups and Lie groups are examples of internal category objects.

We are interested in generalizing this to the context of higher categories. Like always we run into the same problem: A higher category is an infinite tower of data, thus we cannot just list properties the way we did above. We need a tool that allows us to package all the relevant data.

Review of Models of Higher Category Theory

Recall that we studied several models of higher categories:

1. Simplicial Categories
2. Quasi Categories
3. Relative Categories
4. Complete Segal Spaces

In order to build higher categories internal to a higher category, we have to figure out which model lends itself to such generalization.

1. Simplicial Categories don't lend themselves to any such generalization.
2. Quasi-categories could be generalized in any higher category which has notions of simplices and inner horns. Otherwise, we cannot generalize them in an arbitarary higher category.

3. Relative categories can certainly be generalized as they are just categories. We can generalize them by considering category objects with a subcategory object, which represents the weak equivalences. However, it is very often hard to work with them.

4. Complete Segal spaces seem to be generalizable, if not for the single issue that the Reedy fibrancy depends on the existence of a model structure, which we clearly lack in an arbitrary higher category. Ignoring that condition though we can probably figure things and that is exactly what we are going to do.

Segal Objects and their Homotopy Theory

From here on let W be a complete Segal space with finite limits (any model of higher category would work here). We have following definition:

Definition 3.1. Let
$$sW = W^{\Delta^{op}}$$
be the CSS of simplicial objects in W.

Definition 3.2. Let $X : \Delta^{op} \to W$ be a simplicial object. We say X satisfies the *Segal condition* if
$$X_n \to X_1 \times_{X_0} ... \times_{X_0} X_1$$
is a weak equivalence in W.

Definition 3.3. A *Segal object* is a simplicial object in W which satisfies the Segal condition.

We have following basic fact:

Lemma 3.4. *Let X be equivalent to Y in sW. Then X is a Segal object if and only if Y is a Segal object.*

Definition 3.5. Let $Seg(W)$ be the sub CSS of sW consisting of Segal objects. Because of the previous lemma the inclusion map $Seg(W) \to sW$ is (-1)-connected.

Segal objects already have a lot of category theory themselves. Let's study it:

Definition 3.6. Let X be a Segal object. The set of objects in X is the set $map(*, X_0)_0$, where $*$ is the final object.

Definition 3.7. Let X be a Segal object. The set of arrows in X is the set $map(*, X_1)_0$, where $*$ is the final object.

Definition 3.8. For two object $x, y : * \to X_0$, we define the **mapping object** as the following pullback square:

$$\begin{tikzcd}
map_X(x,y) \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner"] & X_1 \arrow[d, "(s,t)"] \\
* \arrow[r] & X_0 \times X_0
\end{tikzcd}$$

Using the Segal condition we can deduce that for three objects $x, y, z : * \to X_0$ we have a have a notion of a "composition". Using similar techniques to that of CSS we can show that this composition map is associative and has an identity map (namely the degeneracy map $s_0 : X_0 \to X_1$).

Let us move on to study equivalences.

Equivalences in Segal Objects

Here we cannot use the definition of equivalences we have used before as it there actually used some properties of the category of spaces. However, there totally is a way to get around it:

Definition 4.1. Let $X_{hoequiv}$ be following pullback:

$$\begin{tikzcd}
X_{hoequiv} \arrow[r, "i"] \arrow[d] \arrow[dr, phantom, "\ulcorner"] & X_3 \arrow[d] \\
X_1 \arrow[r] & X_1^s \times^s_{X_0} X_1^t \times^t_{X_0} X_1
\end{tikzcd}$$

We have following definition:

Definition 4.2. An arrow $* \to X_1$ is called an *equivalence* if it factors through the map $i : X_{hoequiv} \to X_1$.

We have following fact that just follows from the simplicial relations:

Lemma 4.3. *Any map*

$$* \to X_0 \to X_1$$

will factor through $X_{hoequiv}$ i.e. is an equivalence.

Note that the opposite is not necessarily true.

Complete Segal Objects

Now we are in the position to define complete Segal objects:

Definition 5.1. A Segal object X is a *complete Segal object* if the map

$$X_0 \to X_{hoequiv}$$

is an equivalence in the CSS W.

Again we have a similar lemma.

Lemma 5.2. *Let X be equivalent to Y in sW. Then X is a CSO if and only if Y is a CSO.*

Definition 5.3. Let $CSO(W)$ be the subCSS of sW such that all the objects are CSO. Because of the previous lemma the natural inclusion map $CSO(W) \to sW$ is (-1)-connected.

Let us see some examples of these.

Example 5.4. Let x be any object W. Then the constant simplicial object is always a complete Segal object.

Example 5.5. Let W be the CSS of spaces. In this context a complete Segal object would then just be a complete Segal space, up to Reedy fibrancy.

Remark 5.6. This last example might make us think: Is there a way to construct the a homotopically rigorous object without resorting to fibrant replacements?

Example 5.7. Let us see some more interesting example. Let $\mathcal{S}et$ be the category of sets, thought of as a discrete higher category. Then a simplicial object is just a simplicial set. A Segal set is just then a simplicial set that is isomorphic to the nerve of a category, in other words it is just a category. A complete Segal set is actually more restrictive. It corresponds to the nerve of a category that has no nontrivial automorphism. Given the philosophy of complete Segal objects that makes a lot of sense. The homotopy theory of my higher category should be controlled by the zeroth object. But this is now just a set. This means there cannot be any homotopy or in other words, there can't be any non-trivial automorphisms.

Example 5.8. This example raises following question. How can we define a category using the theory of complete Segal objects? The idea is to start with the category of groupoids and look at complete Segal objects in the category of groupoids. But this would then be too general, so we need some restrictions. Exercise: Find the right restrictions.

Example 5.9. Let CSS be the complete Segal space of CSS. Then a CSO in CSS is exactly the higher categorical analogue to a double category i.e. a *higher double category.*

Let's come back to the 1-categorical context and witness one more property of category objects:

Theorem 5.10. *If (O, M) is a category object internal to a category $\mathcal{C}$ and C is an object in $\mathcal{C}$, then then $(Hom_{\mathcal{C}}(C, O), Hom_{\mathcal{C}}(C, M))$ is a category.*

Moreover, we have following simple fact:

Theorem 5.11. *Let $f : C_1 \to C_2$ be a morphism in $\mathcal{C}$. Then we get a functor of categories:*

$$(Hom_{\mathcal{C}}(C_2, O), Hom_{\mathcal{C}}(C_2, M)) \xrightarrow{f^*} (Hom_{\mathcal{C}}(C_1, O), Hom_{\mathcal{C}}(C_1, M))$$

So, essentially we have a functor:

$$(O, M)^* : \mathcal{C}^{op} \to \mathcal{C}at$$

We would like to have an analogous situation for higher categories, but as always things fall apart very fast. As is customary we have to change our view from functors to fibrations to be to handle things appropriately.

Cartesian Fibrations

In order to be able to do that we have to recall Cartesian fibrations.

Definition 6.1. Let X be a simplicial space thought of as a bisimplicial space such that

$$X_{kn} = X_n$$

A map of bisimplicial spaces $Y \to X$ is called a *Cartesian fibration* if it satisfies following three conditions:

1. It is a Reedy right fibration
2. It satisfies the Segal condition
3. It satisfies the completeness condition

Whenever we have fibrations we also have a model structure.

Theorem 6.2. *Let X be a simplicial space. There is a unique, simplicial model structure on $ss\mathcal{S}_{/X}$ such that*

1. *Fibrant objects are the Cartesian fibrations*
2. *Cofibrations are inclusions.*

This is called the Cartesian model structure.

For more details see [We17].

Representable Cartesian Fibrations

Now that we have settled on a model of a map into higher categories, have to characterize those that come from complete Segal objects. Given the construction we have given this is actually quite simple.

Definition 7.1. Let W be a CSS with limits and $x_\bullet$ be a CSO in W. Then the map of bisimplicial spaces, defined level-wise as

$$W_{/x_k} \to W$$

is a Cartesian fibration.

This is exactly the Cartesian fibration that classifies the functor we desired above.

Note that we can determine the "value" of the Cartesian fibation, by looking at the fiber:

$$\begin{tikzcd} map(y, x_k) \arrow[r] \arrow[d] \arrow[dr, phantom, "\ulcorner", very near start] & W_{/x_k} \arrow[d] \\ F(0) \arrow[r, "y"'] & W \end{tikzcd}$$

which is exactly what we were hoping for.

Remark 7.2. Notice how we starting with a completely model independent definition of a complete Segal objects internal to a CSS and managed to embed it into the larger category of Cartesian fibrations, which actually comes with a model structure. This actually allows us to do concrete computations with arbitrary CSO.

This allows us to give following definition:

Definition 7.3. A Cartesian fibration $C \to W$ is called *representable* if there exists a CSO x such that $W_{/x}$ is equivalent to C.

This approach gives us many valuable computational tools to understand complete Segal objects. For example we have following simple Lemma:

Lemma 7.4. *Let x be a simplicial object. Then the map*

$$\Delta_{/k} \to W_{/x_k}$$

over W is a Cartesian equivalence.

This in particular implies that for two CSO x and y the following is a Kan equivalence:

$$Map_{/W}(W_{/x}, W_{/y}) \to Map_{/W}(\Delta_{/\bullet}, W_{/y})$$

A careful analysis of this result gives us a *Cartesian Yoneda embedding*: i.e. there is an equivalence

$$map_{sW}(x, y) \to map_{ss\mathcal{S}_{/W}}(W_{/x}, W_{/y})$$

This is already a good indicator that representable Cartesian fibrations are a good way to study and compare complete Segal objects.

I leave you with the following question:

Let x be a simplicial object in W, (not necessarily a CSO). Then we can still build $W_{/x}$, but it won't be a Cartesian fibration anymore. This means there exists Cartesian equivalence $W_{/x} \to C$ such that $C \to X$ is a Cartesian fibration. Now we might ask ourselves the following: What conditions do we need on W such that we know that C itself is representable? We know this is true in certain cases. For example if W is the CSS of spaces, then a simplicial object is a just a simplicial space and so the Cartesian replacement would just be the fibrant CSS replacement which must exist because we have a model structure.

Something similar can be said about simplicial spaces, but it is unclear where such a result would hold in general.

References

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